

RESULTS REPORT

Malawi Agricultural Commercialisation Dashboard

Maize & Soybean Export Performance Analysis

2020 – 2024 • FAOSTAT Open Data • Microsoft Power BI

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Tool	Microsoft Power BI
Data Source	FAOSTAT — Food and Agriculture Organization of the United Nations
Element Codes	FAO 5910 (Export Quantity) and FAO 5922 (Export Value)
Period	2020 to 2024

208,964 t	\$218.9M	84.5%	\$108.1M
Total Export Volume	Total Revenue (USD)	Soybean Volume Share	Peak Soybean Revenue

EXECUTIVE SUMMARY

Project Overview

This report presents the results of a data-driven analysis of Malawi's agricultural export performance for maize and soybean over the five-year period from 2020 to 2024. Using open data published by the Food and Agriculture Organization of the United Nations (FAOSTAT), the analysis tracks export volumes in tonnes, export revenues in USD, implied market prices per tonne, and year-on-year growth rates for each crop. The dashboard was built entirely in Microsoft Power BI.

Key Findings

- 1

Soybean dominates both volume and revenue.

Across the full five years, soybean accounted for 84.5% of combined export volume and 88.5% of combined revenue. This confirms soybean as Malawi's structurally central agricultural export commodity.
- 2

The 2021 revenue figure is an outlier, not a baseline.

Soybean generated \$108.1M in 2021 — nearly three times any other year. This was driven by a global commodity price spike from \$478/tonne in 2020 to \$2,358/tonne in 2021. Excluding 2021, average annual soybean revenue was approximately \$21.3M.
- 3

Maize exports are constrained by domestic food security.

Maize volumes fluctuated from 634 tonnes in 2020 to a peak of 20,257 tonnes in 2022, then fell to 357 tonnes in 2024. This extreme volatility reflects maize's dual role as a food security and export crop.
- 4

2023 demonstrated soybean's real production ceiling.

The 2023 soybean export volume of 57,897 tonnes — the highest in the dataset — was achieved at a normalised price of \$696/tonne without any global price windfall. This is the strongest indicator of underlying export capacity.
- 5

Volume and price moved inversely for maize.

In 2022, when maize hit peak volume, the implied price fell to its lowest at \$524/tonne. In years with lower volumes, prices exceeded \$1,100/tonne — suggesting surplus-driven rather than commercially managed exports.

SECTION 1 — KPI DASHBOARD

Key Performance Indicators

The KPI page presents five headline metrics derived from the full dataset using DAX measures in Microsoft Power BI. These figures represent five-year combined totals and records across both crops. A Year slicer allows filtering all metrics to any single year from 2020 to 2024.

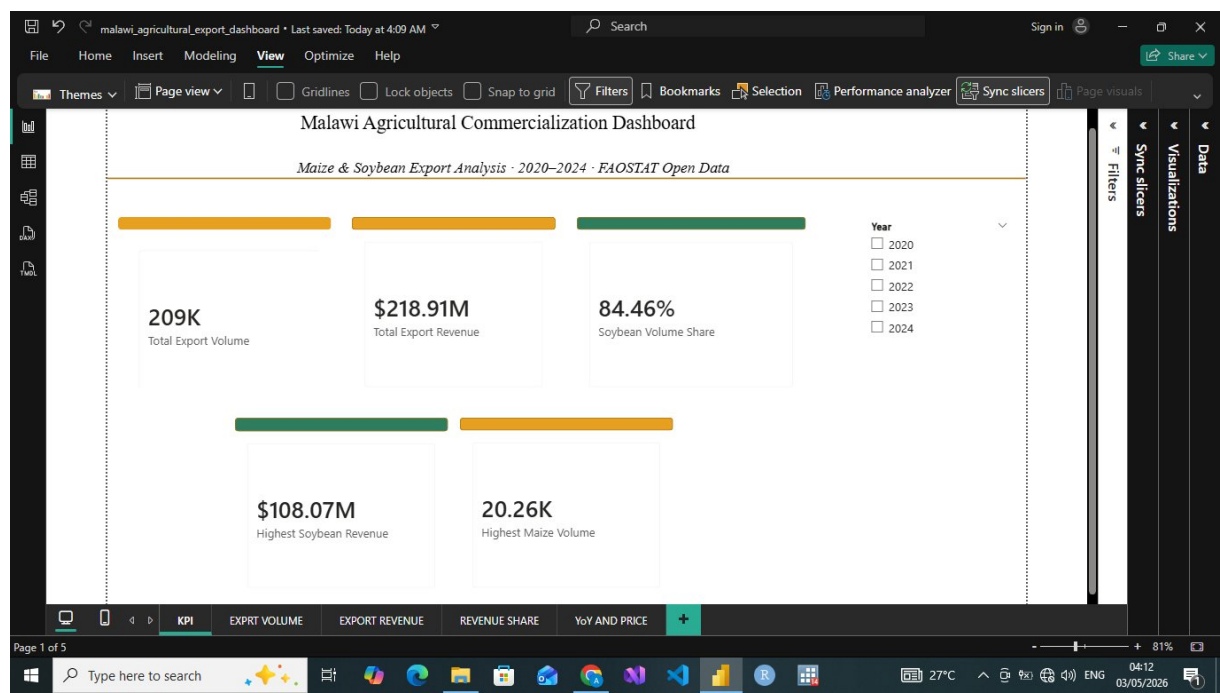


Figure 1 — KPI Dashboard Page. Five-year totals and peak records for maize and soybean exports.

KPI Metrics Explained

Metric	Value	Interpretation
Total Export Volume	209K tonnes	Combined maize and soybean exports across 2020-2024
Total Export Revenue	\$218.91M	Total foreign exchange earned from both crops
Soybean Volume Share	84.46%	Soybean's proportion of combined 5-year export volume
Highest Soybean Revenue	\$108.07M	Peak year for soybean — 2021 global commodity price spike
Highest Maize Volume	20.26K tonnes	Peak maize year — 2022 surplus-driven exports

The Year slicer on this page (top right, dropdown style) filters all five KPI cards simultaneously when a year is selected.

SECTION 2 — EXPORT VOLUME ANALYSIS

Export Volume by Crop

The export volume page shows two visuals: a clustered horizontal bar chart comparing maize and soybean volumes side by side for each year, and a stacked 100% bar chart showing each crop's share of the combined total. Together they establish soybean's dominance while highlighting 2022 as the only year where maize held a meaningfully large share.

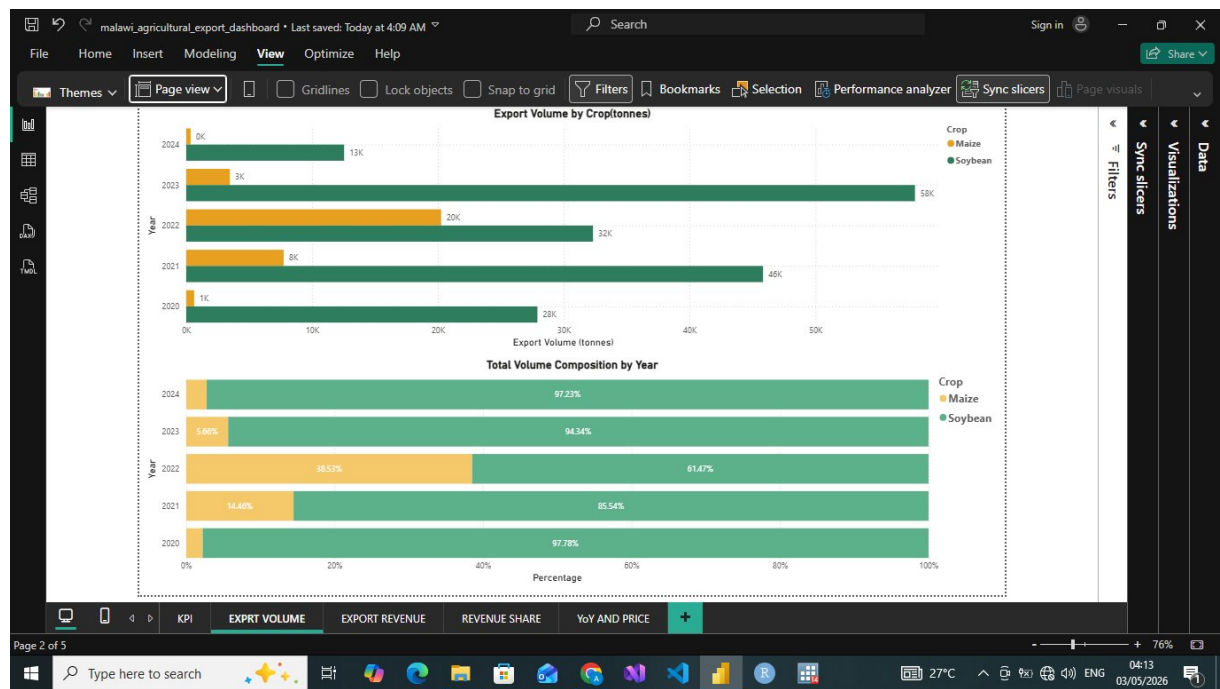


Figure 2 — Export Volume by Crop (top) and Total Volume Composition by Year (bottom). Source: FAOSTAT Code 5910.

Volume Data Table

Year	Maize Volume (t)	Soybean Volume (t)	Combined (t)	Soybean Share
2020	634	27,903	28,537	97.78%
2021	7,748	45,829	53,577	85.54%
2022	20,257	32,320	52,577	61.47%
2023	3,472	57,897	61,369	94.34%
2024	357	12,548	12,905	97.23%
TOTAL	32,468	176,497	208,965	84.5% avg

Key Observations

- Soybean held over 97% of combined exports in 2020, 2024 — the two smallest maize years.
- 2022 was the only year maize approached significance, reaching 38.5% of combined volume.
- Soybean's 2023 volume of 57,897 tonnes was the highest single-year figure for either crop.

- Maize's volume range (19,900 t) is extreme relative to its mean, confirming high volatility.

SECTION 3 — EXPORT REVENUE ANALYSIS

Export Revenue by Crop

The export revenue page presents a clustered bar chart of annual revenue in millions of USD for each crop, and a stacked percentage bar chart showing each crop's revenue share per year. The 2021 soybean bar — at \$108M — extends to nearly the full chart width, dwarfing every other bar and making the price spike immediately visible.

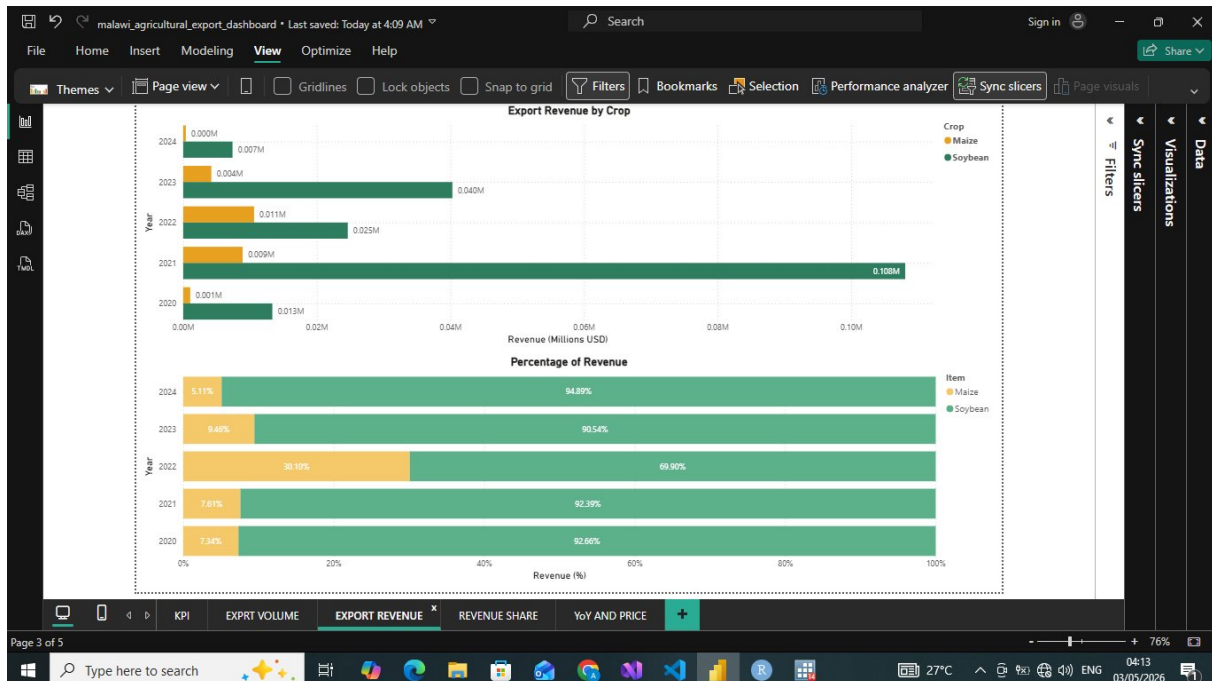


Figure 3 — Export Revenue by Crop (top) and Percentage of Revenue by Year (bottom). Source: FAOSTAT Code 5922.

Revenue Data Table

Year	Maize Revenue	Soybean Revenue	Total Revenue	Soybean Share
2020	\$1.1M	\$13.3M	\$14.4M	92.66%
2021	\$8.9M	\$108.1M	\$117.0M	92.39%
2022	\$10.6M	\$24.6M	\$35.2M	69.90%
2023	\$4.2M	\$40.3M	\$44.5M	90.54%
2024	\$0.4M	\$7.4M	\$7.8M	94.89%
TOTAL	\$25.2M	\$193.7M	\$218.9M	88.5% avg

The 2021 soybean revenue of \$108.1M was driven by an exceptional global price spike — the implied price per tonne rose from \$478 in 2020 to \$2,358 in 2021, a nearly fivefold increase. Excluding 2021, average annual soybean revenue across the remaining four years was \$21.4M — a more realistic planning baseline.

SECTION 4 — REVENUE & VOLUME SHARE

5-Year Revenue and Volume Share

Two donut charts display the five-year cumulative split between maize and soybean — one for total revenue and one for total volume. These confirm soybean’s structural dominance and show that its revenue share (88.5%) slightly exceeds its volume share (84.5%), indicating a higher average unit value over the period.

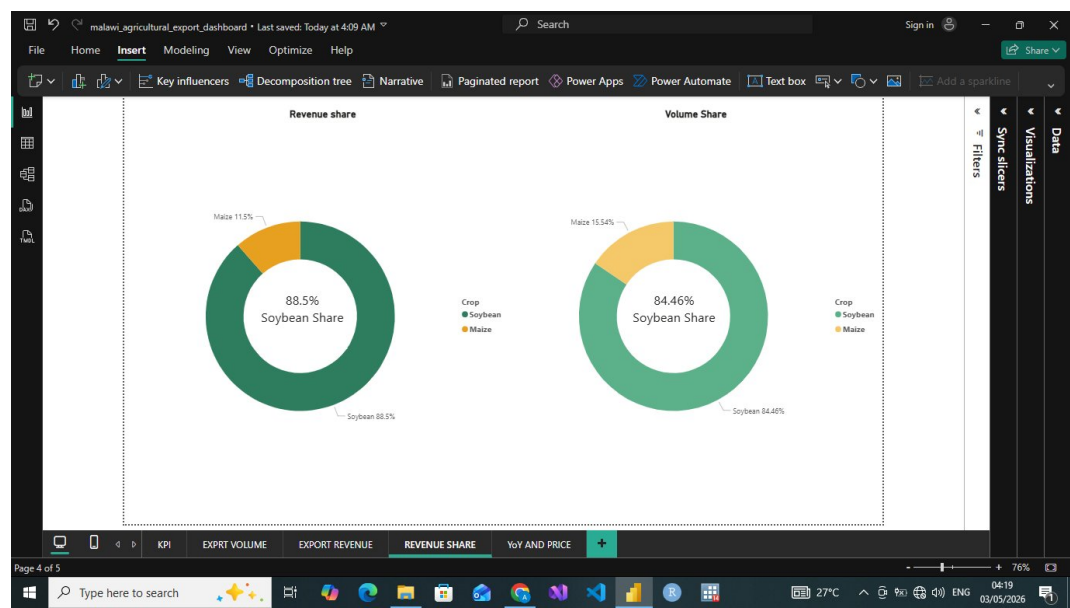


Figure 4 — 5-Year Revenue Share (left) and Volume Share (right). Soybean in green, Maize in amber.

Crop	Volume Share	Revenue Share	Observation
Soybean	84.46%	88.5%	Revenue share exceeds volume share — higher average unit
Maize	15.54%	11.5%	Revenue share below volume share — lower average unit price

SECTION 5 — PRICE & YEAR-ON-YEAR GROWTH

Implied Price per Tonne and YoY Volume Growth

The final dashboard page places two analytical charts side by side: a line chart tracking the implied export price per tonne for each crop (revenue divided by volume), and a clustered column chart showing year-on-year percentage changes in export volume. The line chart captures the dramatic 2021 price spike for soybean; the column chart reveals the extreme growth swings that characterise maize in particular.

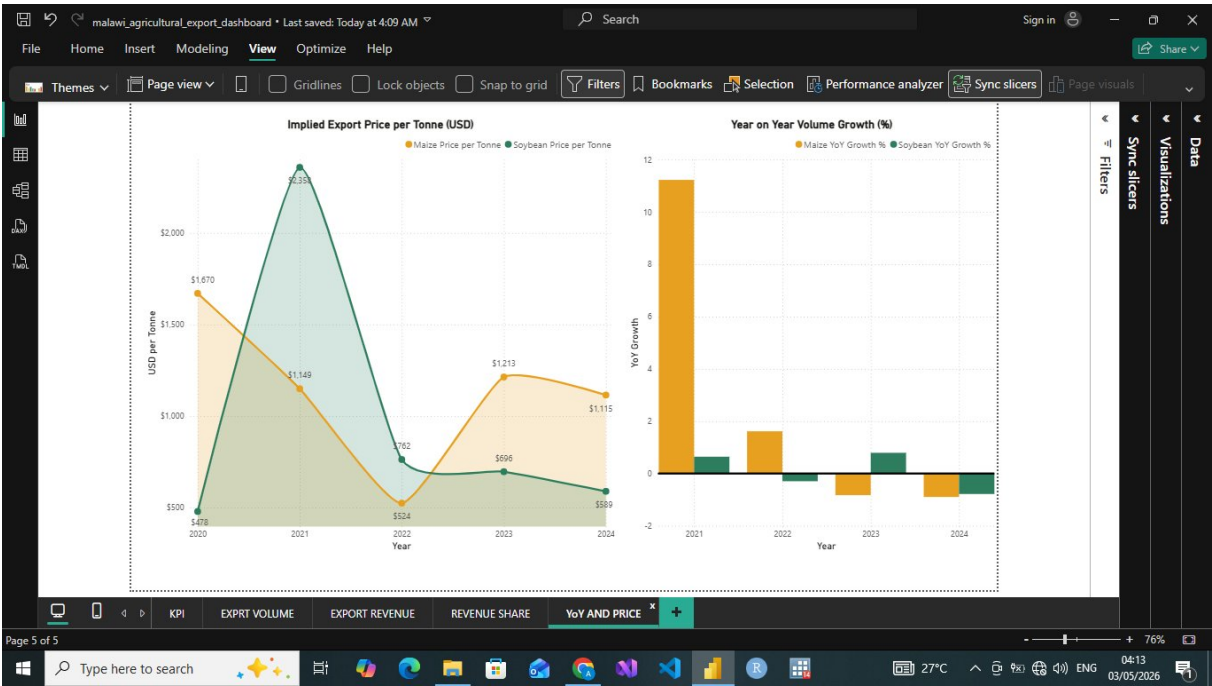


Figure 5 — Implied Export Price per Tonne (left) and Year-on-Year Volume Growth % (right).

Price per Tonne Table

Year	Maize (USD/t)	Soybean (USD/t)	Key observation
2020	\$1,670	\$478	Soybean price low — pandemic market disruption
2021	\$1,149	\$2,358	Soybean spike — global commodity boom
2022	\$524	\$762	Maize price at 5-year low despite peak volume
2023	\$1,213	\$696	Maize price recovers; soybean normalised
2024	\$1,115	\$589	Both prices declining; likely partial-year data
5yr Avg	\$1,134	\$977	Maize 5yr average higher than soybean average

Year-on-Year Growth

Maize's 2021 bar stands at approximately +1,123% — the largest single-year jump in the dataset, from a very low 2020 base of 634 tonnes. Soybean grew 64% in 2021, contracted 30% in 2022, then recovered 79% in 2023. Both crops show negative growth in 2024, consistent with the partial-year data caveat throughout this

report.

SECTION 6 — CONCLUSIONS

Conclusions

1. Soybean is the anchor of Malawi's agricultural export strategy.

With 84.5% of volume and 88.5% of revenue across five years, soybean is the defining crop for agricultural export earnings. Policy, infrastructure investment, and agronomic support should concentrate here to maximise foreign exchange returns.

2. The 2021 windfall must not distort planning.

A single year generated nearly half of the five-year total soybean revenue. The normalised baseline of approximately \$20-40M per year is the realistic planning range.

3. Maize commercialisation is fundamentally constrained.

Maize's food security role means export volumes will always be secondary to domestic demand. The priority should be improving maize's implied export price — which averaged \$1,134/tonne — by targeting premium markets and improving post-harvest quality.

4. 2023 sets the production target for soybean.

57,897 tonnes at \$696/tonne generating \$40.3M in a non-windfall year is the metric the soybean sector should aim to sustain and surpass going forward.

5. Full 2024 data should be incorporated when available.

The sharp declines in 2024 may reflect incomplete FAO reporting. The analysis should be refreshed once full 2024 FAOSTAT data is published before drawing structural conclusions.

Data Reference

FAOSTAT. Trade — Crops and Livestock Products. Food and Agriculture Organization of the United Nations. Element Code 5910: Export Quantity (tonnes). Element Code 5922: Export Value (1,000 USD). Country: Malawi. Years: 2020–2024.